Supplementary material for Wind, slope, and the want of a tail: a computational feasibility study of Eilmer of Malmesbury's flight

This supplement gives the source extracts behind the historical and biomechanical inputs (S1), the parameter ranges sampled in the Monte Carlo (S2), the ridge-lift field (S3), and worked answers to the exercises of Section 6 of the main text (S4). Figure, table, and equation numbers without an S refer to the main text.

S1. Source extracts

S1.1 The flight, the distance, the injury, and the tail. The account is one passage of William of Malmesbury, *Gesta Regum Anglorum* (book II, ch. 225), in the Oxford Medieval Texts edition [1]:

> *Is erat litteris, quantum ad id temporis, bene imbutus, aeuo maturus, immanem audaciam prima iuuentute conatus: nam pennas manibus et pedibus haud scio qua innexuerat arte, ut Daedali more uolaret, fabulam pro uero amplexus, collectaque e summo turris aura, spatio stadii et plus uolauit; sed uenti et turbinis uiolentia, simul et temerarii facti conscientia, tremulus cecidit, perpetuo post haec debilis, et crura effractus. Ipse ferebat causam ruinae quod caudam in posteriori parte oblitus fuerit.*

In the standard English rendering:

> He had by some means, I scarcely know what, fastened wings to his hands and feet so that, mistaking fable for truth, he might fly like Daedalus, and, collecting the breeze on the summit of a tower, he flew for more than the distance of a furlong. But, agitated by the violence of the wind and the swirling of air, as well as by awareness of his rashness, he fell, broke his legs, and was lame ever after. He himself used to say that the cause of his failure was his forgetting to put a tail on the back part.

The distance is a *stadium*, a furlong, about 200 m.

S1.2 The tibia fracture threshold. The 7.5–11.3 kN per-leg range in Fig. 8 follows a systematic review [2], which reports that

> tibial fracture thresholds range from ∼7.5 kN in female specimens to 11.3 kN under combined axial and bending loads, with fibula contribution increasing axial tolerance by ∼10%.

The upper figure traces to Salzar, Lievers, and Bailey [3], who recommend a tibial axial threshold of about 10.3 kN, rising to about 11.3 kN with the fibula.

S1.3 The medieval wind regime. Trouet et al. [4] report

> a 947-year-long multidecadal North Atlantic Oscillation (NAO) reconstruction [with] a persistent positive NAO during the MCA [Medieval Climate Anomaly].

Ortega et al. [5] revise this, finding

> no persistent positive NAO during the medieval period, but [suggesting] that positive phases were dominant during the thirteenth and fourteenth centuries.

Neither reconstruction indicates the weak westerlies of a persistent negative NAO in the eleventh century, so the modern wind statistics are not taken to overstate Eilmer's. The analysis does not assume that medieval winds were stronger.

S2. Sampled parameter ranges

Table S1: Ranges sampled in the Monte Carlo (Fig. 7). All are uniform. The glide ratio is not sampled; it follows from aspect ratio, span efficiency, and zero-lift drag through Eq. (2) and spans 3.1–7.9. The launch wind is drawn from the ERA5 southwest-sector hours at 3–16 $\mathrm{m\,s^{-1}}$.

Parameter	Range	Basis
Wing area	5–10 $\mathrm{m^2}$	wing spanning hands to feet
All-up mass	70–90 kg	adult man and a timber frame
Trim lift coefficient C_L	0.7–1.1	cambered membrane [6]
Aspect ratio A	3–5	span of one to two arm spans
Span efficiency e	0.6–0.8	untwisted, non-elliptical wing [7]
Zero-lift drag C_{D0}	0.05–0.15	membrane, prone body, frame [8, 9]
Tower height	15–25 m	around White's 18 m [10]
Slope angle θ_s	8°–18°	DEM 90th percentile as a floor
Lift efficiency η	0.5–1.0	flow following the surface
CG x_{cg}	0.35–0.55 c	hands at leading edge, feet at trailing edge
Neutral-point shift from a tail	0.18–0.32 c	tail size

S3. Ridge-lift field

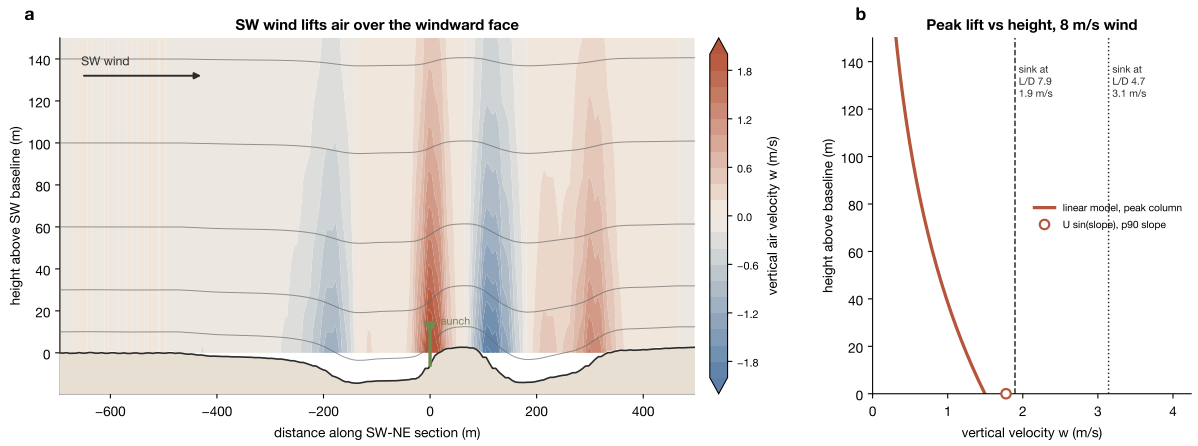


Figure S1: Ridge lift over the abbey spur. The linear model of Eq. (1) over the measured southwest–northeast section. **(a)** Vertical air velocity and streamlines in an $8\,\mathrm{m\,s^{-1}}$ southwest wind. **(b)** Vertical air velocity against height above the point of strongest lift, $1.5\,\mathrm{m\,s^{-1}}$ at the surface. The open circle is the surface estimate $U\sin\theta_s$ used in the glide model at the 90th-percentile slope. The vertical lines are the still-air sink of the nominal craft at the highest and central glide ratios.

S4. Answers to the exercises

1. Energy at launch. Energy conservation gives $\Delta h = (V^2 - v_0^2)/2g = 11.5\,\mathrm{m}$, most of an 18 m tower. A headwind of $V - v_0 \approx 13.1\,\mathrm{m\,s^{-1}}$ gives the full airspeed at release. The steady-glide estimate for the best wing is $H\,(L/D) \approx 142\,\mathrm{m}$; integrating Eq. (3) gives 14 m, because the craft spends its height on the dive and the pull-out.

2. Glide ratio. Minimising C_D/C_L with $C_D = C_{D0} + C_L^2/(\pi e A)$ gives $C_L^2 = \pi e A C_{D0}$, at which the two drag terms are equal and $L/D = \frac{1}{2}\sqrt{\pi e A/C_{D0}}$. The corners of Table S1 give 3.1–7.9. C_{D0} varies by a factor of three and moves L/D by $\sqrt{3}$, more than A or e; a drag measurement on a model of the wing and body would narrow it most.

3. Velocity triangles. Heading into the wind, the air and wind velocities are collinear and the ground speed is $V - U$. Crabbing along a track perpendicular to the wind, the heading component into the wind must equal U, leaving $\sqrt{V^2 - U^2}$ along the track. The headwind adds to the airspeed at release on the first track, but subtracts from the ground speed for the whole flight.

4. Ridge lift and sink. The sink rates are $3.1\,\mathrm{m\,s^{-1}}$ at L/D 4.7 and $1.9\,\mathrm{m\,s^{-1}}$ at L/D 7.9. On a slope of 12.8° the lift matches them at $U \approx 14.2\,\mathrm{m\,s^{-1}}$ and $8.5\,\mathrm{m\,s^{-1}}$. Southwest winds of $8\,\mathrm{m\,s^{-1}}$ or more occur in 3.4 % of hours. The full simulation needs less wind ($5.7\,\mathrm{m\,s^{-1}}$ for the best wing) because the tower height covers part of the sink.

5. Pitch stability. With state (α, q), the matrix $\begin{pmatrix} Z_\alpha & 1 \\ M_\alpha & M_q \end{pmatrix}$ has a positive real eigenvalue when its determinant $Z_\alpha M_q - M_\alpha$ is negative, that is, when $M_\alpha > Z_\alpha M_q$. Since $M_\alpha \propto x_{\mathrm{cg}} - x_{\mathrm{np}}$ and $Z_\alpha M_q > 0$, divergence begins slightly aft of the neutral point, at $0.30\,c$ for the nominal craft. At $x_{\mathrm{cg}} = 0.45$, $t_2 = 0.24\,\mathrm{s}$. A ruler-drop test usually gives a simple reaction time of about 0.2 s, so the disturbance doubles about once per reaction.

6. Monte Carlo. For $k = 2\,748$ and $n = 40\,000$, $\hat{p} = 6.9\,\%$ and the Wilson half-width is close to $1.96\sqrt{\hat{p}(1-\hat{p})/n} \approx 0.25\,\%$, giving 6.6–7.1 %. Another seed changes the fraction by about this amount.

7. Impact. The onset speeds are 6.1–14.1 $\mathrm{m\,s^{-1}}$. At $V = 15.1\,\mathrm{m\,s^{-1}}$ and 50°, the vertical speed is $11.6\,\mathrm{m\,s^{-1}}$, above the onset speed for the central stopping distance. A glide into the ground at the sink rate, 1.9–3.1 $\mathrm{m\,s^{-1}}$, is not. The injury implies a steep, fast arrival, consistent with a loss of control but not proof of one.

References

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